Corrected MCQs

1. **B.** The sign $f'' > 0$ means that f' It does not give the sign of convexity. f' or the f.
2. **C.** The derivative of $e^{u(x)}$ is $u'(x)e^{u(x)}$. Here $u(x) = 1 + x^2$, therefore $u'(x) = 2x$. Forgetting this factor is a miscomposition.
3. **B.** A deviable convex function is above each of its tangents on the interval of
   It is not necessary for these tangents to be horizontal.
4. **B.** The positive sign of f' then negative gives growth and then decrease around a : maximum local, and not minimum.
5. **C.** $|x| \to 0$ when $x \to 0$, so continuity. The slopes to the left and right are worth -1 and 1, therefore not Differentiability in 0.
6. **A.** The VIT uses continuity throughout the interval. No null derivative is required; an interval is required. reduced does not give a change of sign.

7. **B.** Continuity and change of sign ensure existence. A function may have several roots: uniqueness requires an additional argument.
8. **B.** A strictly monotonous function is injective: a given value has at most one preimage. Being bounded or paired does not insure this property.
9. **B.** As long as no exact root is found, the sub-interval with the bounds is retained. images of opposite signs; its length is divided by two.
10. **B.** Over an interval where $f'' < 0$, the derivative of f' is negative, so f' His sign is still in study.

True/false. False: e^x Never cancels. True if f is derivable over the interval: it is a characterization
False: an approaching number alone does not prove existence; dichotomy must start
a time interval where continuity and change of sign are established.

Corrected applications and support

Application 1. g is defined on $[1/2;+\infty[$ and drawable on $]1/2;+\infty[$, $g'(x) = 1/\sqrt{2x-1} > 0$. h is derivable on $\mathbb{R}$, $h'(x) = -2xe^{1-x^2}$.

Application 2. The inner derivative is 3, so $f' = 12(3x+1)^3$. The coefficient x in the development $(1+3x)^4$ is 12, where $f'(0) = 12$, incompatible with the proposed formula.

Implementation 3. C is f, A is f' and B is f''. The zeros of A in ± 1 correspond to the extrema of C; the zero of B in 0 separates the decay and growth of A. Calculation: $f' = 3x^2 - 3$, $f'' = 6x$.

Implementation 4. The three are continuous. The polynomials are derivable; $|x|$ a side slopes -1 1. For x^4, $f'' = 12x^2 \geq 0$, therefore no inflection in 0; for x^3, $f'' = 6x$ change of sign, so 0 inflection. $|x|$ is convex on $\mathbb{R}$ and does not change convexity to 0.

Application 5. The staircase function never takes the value 0: it is not continuous in 0. x^2 is continuous but is not strictly monotonous on $\mathbb{R}$; $x^2 = 1$ a both solutions ± 1.

Implementation 6. Lay down $G(x) = e^{-x} - x$, continue, with $G(0) = 1 > 0$ and $G(1) = e^{-1} - 1 < 0$. $G' = -e^{-x} - 1 < 0$ ensures uniqueness. $G(1/2) > 0$, then $G(3/4) < 0$: interval $[1/2;3/4]$. Since $[0;1]$, seven steps give a width 1/128; six are not enough.

Guidance and review

Hide the formulas during the first work on the three curves. If the student confuses signs and change, point a portion of f negative but increasing. After a week, ask a new justification of uniqueness without the "use VIT" instruction.

Conduct and correction

Benchmarks for the teacher

The progression concentrates this chapter in three sessions: really divide S29 between diversions and convexity, then repeat the evidence of the tangents in S33. Continuous/derivable/convex vocabulary It should be distinguished by examples. Continuity gives existence by the VIT; strict monotonicity gives unity. For dichotomy, require an initial interval where the sign changes and a continuous function. A change of convexity, not just the cancellation of f'', characterizes a deflection point.

Corrected activity

1. $f'(x) = 3x^2 - 3 = 3(x-1)(x+1)$ and $f''(x) = 6x$. The sign of f' is positive on $]-\infty, -1[$ and $]1, +\infty[$, zero in -1, 1, negative between these values. f'' is negative to the left of 0, zero to 0, positive to Right.

2. Variations and limitations are summarized by

x	$-\infty$		-1		1		$+\infty$
$f'(x)$		$+$	0	$-$	0	$+$	
$f(x)$	$-\infty$	$\nearrow$	2	$\searrow$	-2	$\nearrow$	$+\infty$

   The sign of f'' gives concavity on $]-\infty, 0]$ and convexity on $[0, +\infty[$.

3. Convexity changes to 0, and $f(0) = 0$: the point of inflection is $(0, 0)$. Its tangent is an equation $y = f(0) + f'(0)x = -3x$.

4. The function is continuous. $]-\infty, -1]$, it is strictly increasing $-\infty$ to 2: level 1 is reached exactly once. $[-1, 1]$, it strictly decreases from 2 to -2 : a second solution. $[1, +\infty[$, it grows strictly from -2 à $+\infty$: a third. Points -1, 1 are not no solutions, so no solution is counted twice.

5. Continuity provides the existence of a preimage for intermediate values. not unique: the function studied is continuous and level 1 has three preimage.

E1. For $u = 2x - 1$, $(u^5)' = 5u'u^4 = 10(2x-1)^4$. For $v = -x^2$, $(e^v)' = v'e^v = -2xe^{-x^2}$. Both functions are derivatable on $\mathbb{R}$.

E2. $f' = 4x^3 - 12x$ then $f'' = 12(x^2 - 1)$. The sign of f'' is positive outside of $[-1, 1]$negative inside. f is convex on $]-\infty, -1]$ and $[1, +\infty[$, concave on $[-1, 1]$. Changes in convexity give inflections $(-1, -5)$ and $(1, -5)$.

E3. The tangent is $y = f(1) + f'(1)(x-1) = e + e(x-1) = ex$. Since $f''(x) = e^x > 0$, the exponential is convex and its curve is above this tangent: $e^x \geq ex$ on $\mathbb{R}$.

E4. Poisons $g(x) = x^3 + 2x - 5$. This polynomial is continuous and $g'(x) = 3x^2 + 2 > 0$ on $\mathbb{R}$. In addition $g(1) = -2 < 0$ and $g(2) = 7 > 0$: the VIT gives a zero in $]1, 2[$, and strict growth on $\mathbb{R}$ insurance that it is unique.

E5. After n steps, length is $\frac{2}{2^n}$. The requested condition is equivalent to $2^n \geq 2000$. Like $2^{10} = 1024 < 2000 \leq 2048 = 2^{11}$, eleven steps are enough and ten steps are not enough.

Corrected from DM4

1. $f(0) = 1$ and $\lim_{x\to 0^+} f(x) = 1$. In $+\infty$, $f(x) = xe^{-x} + e^{-x} \to 0$ by comparative growth. product, $f'(x) = e^{-x} - (x+1)e^{-x} = -xe^{-x}$. This derivative is strictly negative for $x > 0 : f$ decrease strictly from 1 to 0 on $[0, +\infty[$.

2. $f''(x) = -e^{-x} + xe^{-x} = (x-1)e^{-x}$. So f is concave on $[0,1]$, convex $[1, +\infty[$. The change of convexity to 1 gives the point of inflection $(1, \frac{2}{e})$.

3. f is continuous and strictly decreasing on $[0, +\infty[$; $f(0) = 1 > \frac{1}{2}$ and its limit 0 is lower to $\frac{1}{2}$. The corollary of the VIT gives a single solution α. In addition, $f(1) = \frac{2}{e} \simeq 0{,}7358 > \frac{1}{2}$ and $f(2) = \frac{3}{e^2} \simeq 0{,}4060 < \frac{1}{2}$, therefore $1 < \alpha < 2$.

4.
```
From math import exp

def root_dm4():
    a, b = 0.0, 2.0
    while b-a > 1e-3:
        m = (a+b)/2
        if (m+1)*exp(-m) > 0.5:
            a = m
        Else :
            b = m
    return a, b
```

   The decline keeps $f(a) \geq \frac{1}{2} \geq f(b)$; α remains in $[a,b]$. After eleven steps, we get $[1{,}677734375, 1{,}6787109375]$, length $\frac{1}{1024} < 10^{-3}$. Value $\alpha \simeq 1{,}67835$ is compatible with This framework.

5. The tangent in 0 is $y = f(0) + f'(0)x = 1$. Variations $f(x) \leq 1$, so the whole curve is below this tangent. This does not characterize concavity: it should be below of all the tangents, while f becomes convex after 1.

Corrected work in the manual or an anal

For $f(x) = 5x^2e^x$, the derivation of a product gives

$$f'(x) = 10xe^x + 5x^2e^x = 5x(x+2)e^x.$$

Since $e^x > 0$, f' is positive on $]-\infty;-2[$ and $]0;+\infty[$negative on $]-2;0[$, none in -2 and 0. function thus grows up to $f(-2) = 20e^{-2}$, decreases until $f(0) = 0$, then grows. Its limits are 0 in $-\infty$ (combined growth) and $+\infty$ in $+\infty$. A new diversion gives

$$f''(x) = 5(x^2+4x+2)e^x = 5\big((x+2)^2-2\big)e^x.$$

Poisons $a = -2-\sqrt{2}$ and $b = -2+\sqrt{2}$. The polynomial factor is positive outside of $[a;b]$negative in $]a;b[$. Thus f is convex on $]-\infty;a]$ and $[b;+\infty[$, concave on $[a;b]$. The sign changes to both roots: the points of inflection are $(a;5a^2e^a)$ and $(b;5b^2e^b)$.

The table of variations is summarized as follows:

x	$-\infty$		-2		0		$+\infty$
$f'(x)$		$+$	0	$-$	0	$+$	
$f(x)$	0	$\nearrow$	$20e^{-2}$	$\searrow$	0	$\nearrow$	$+\infty$

For question 5, the window below makes visible the two changes of convexity and both extrema. Marked points correspond to calculated inflection points; the plot is used to control consistency of the study.

